BADR OBTEL

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EDUCATION

American University of the Middle East (AUM) **Egaila, Kuwait**
Bachelor of Science in Electrical Engineering 2024 – 2028

- GPA: 3.79/4.0 | Dean's Honors List (Freshman & Sophomore Year)
- Robotics Club: Tear-down and reassembly of multirotor drones and ground robots; analyzed power distribution, motor drivers, ESCs, and embedded control.
- Coursework: Digital System Design, Linear Circuit Analysis, Electronic Measurement Tech. Lab, Electronic Analysis & Design, Linear Algebra, Calculus I–III

TECHNICAL SKILLS

- **Embedded & Hardware:** ESP32, STM32, Arduino, PCB Design (KiCAD), Power Regulation
- **Digital Design & Silicon:** Verilog, RTL-to-GDSII flow, Yosys, OpenROAD/LibreLane, Magic VLSI, KLayout, SKY130 PDK
- **Programming:** C, Python, Verilog
- **Tools:** Circuit Simulation (OrCAD, LTSpice), 3D CAD (Fusion360), VS Code, Computer Vision
- **Cloud & DevOps:** Supabase, Azure DevOps, Linux, Git

EXPERIENCE

Lear Corporation | E-Systems **July 2026 – Aug 2026**
Software Engineering Intern, Embedded DevOps
Build and release engineering for automotive electronic-systems validation

- Re-architected a PowerShell hardware-in-the-loop test runner to execute cases in parallel rather than sequentially, eliminating a hang that stalled validation runs for up to 5.5 hours and cutting a full pass to roughly 5 minutes.
- Independently designed and built CONDUIT, a local-first release-control dashboard unifying Azure DevOps with Palantir Foundry to auto-summarize deployment logs and replace a multi-page manual log-review workflow; adopted by full-time engineers.
- Debugged failing LIN-bus lighting and engine-cranking security-access test scripts across the CD_TEST release pipeline, clearing the large majority of logged errors and restoring clean sign-off builds.
- Identified and patched a latent CI gate defect that passed pull requests with zero linked work items, closing a false-positive path that let unverified changes report as successful.

PROJECTS

CLEAVE *RV32I Single-Cycle Core to SKY130 GDSII* *Verilog, Yosys, OpenROAD/LibreLane, Magic, sky130*
From-scratch RISC-V core hardened through a full open-source RTL-to-GDSII flow

- Designed a from-scratch RV32I single-cycle processor core in Verilog implementing the full base integer ISA across a modular datapath: PC, 32x32 register file, ALU, immediate generator, branch comparator, control unit, and byte-addressable data memory.
- Hardened the RTL to a DRC/LVS-clean sky130 GDSII through a full open-source flow (Yosys synthesis; OpenROAD/LibreLane place-and-route and CTS; Magic/KLayout/netgen sign-off) at a 25 MHz target, packaged as a TinyTapeout 1x1 tile.
- Verified every module and the integrated core with 10 self-checking Icarus Verilog benches (9 unit plus a full-ISA integration proof), validating ALU ops, store/load round-trips, all branch conditions, and JAL/JALR against hand-computed values through a debug port.

NERONA *STM32N6 Computer Vision Edge Platform* *KiCAD, STM32N6, MIPI CSI-2, STM32CubeMX*
Custom PCB targeting the STM32N6 NPU for on-device CV inference with no cloud dependency

- Designed PCB power delivery, MIPI CSI-2 camera interface, and memory subsystem around the STM32N6 NPU for accelerated on-device CV inference.
- Completed schematic and layout in KiCAD under signal-integrity and EMC rules, and built the deployment path for quantized object-detection models onto the NPU via STM32CubeMX.

AZIMUTH *ESP32-S3 Multi-Sensor Navigation Board* *KiCAD, ESP32-S3, GNSS, IMU, Sensor Fusion*
Custom navigation board with defined sensor roles and redundancy planning

- Designed a navigation board assigning each sensor a defined role: GNSS for position, magnetometer for heading, IMU accelerometer for tilt correction, and gyroscope for smoothing.
- Planned IMU dead-reckoning as a fallback for continuous heading and motion estimation during GNSS dropout, so navigation degrades gracefully rather than failing on single-sensor loss.

LANGUAGES & SOFT SKILLS

- **Languages:** English (Native) | Arabic (Fluent: Darija; Conversational: Kuwaiti & Egyptian) | French (Beginner)
- **Leadership:** Acted as de-facto lead across university team projects, driving technical execution and delivering final presentations, reports, and demos.